

DATA DESCRIPTION PAPER

A multi-source dataset of CDF projects, budgets, skills-development records, and integrated development planning data from Mufulira Municipal Council, Zambia

Kasonkomona Mulenga - **2021519929**, Ephetred Ndhlovu - **2021463681**,
Kalebalika Chileshe - **2021452344**, Mambwe Luke Chilebela - **2021480348**, Moses Kaluba -
2021387283

University of Zambia / School of Natural Sciences / Department of Computer Science

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Abstract

This data article describes a curated multi-source dataset assembled from publicly accessible digital records published by Mufulira Municipal Council in Zambia. The data cover four principal areas: Constituency Development Fund (CDF) community projects, approved municipal budgets and revenue streams, CDF skills-development bursary records, and governance and Integrated Development Plan (IDP) information. Source material included council webpages and official PDF documents. Digital PDF tables were extracted with pdfplumber, while scanned documents were converted to images with pdf2image and processed using Tesseract optical character recognition through pytesseract. Requests and BeautifulSoup were used to retrieve webpages and discover relevant PDF links, regular expressions and predefined reference lists were used to identify selected fields, and Pandas was used to structure the extracted records. The resulting files are stored as pipe-delimited CSV data to support reproducible analysis. The collection contains project, financial, demographic, health facility, public-consultation, and capital-investment attributes and is intended to support data mining, public-finance analysis, local-development research, and comparative analysis of local government records in Zambia.

Keywords

Local government; CDF; municipal budgets; public finance; development planning; Zambia; open data

Specifications Table

Specification	Description
Subject	Data mining; public administration; local government; public finance
Specific subject area	Machine-readable local-authority data describing development projects, municipal budgets and revenue, bursary records, administrative demographics, health facilities, public-consultation issues, and planned capital investments.
Type of data	Pipe-delimited CSV files; extracted tabular data; text fields; numeric financial and demographic fields.
How the data were acquired	Official Mufulira Municipal Council webpages and PDF documents accessed with Python Requests; HTML link discovery with BeautifulSoup; digital table extraction with pdfplumber; OCR of scanned pages using pdf2image and Tesseract/pytesseract; structuring with Pandas.
Data format	Raw and cleaned/structured CSV datasets using the pipe character () as the delimiter.
Parameters for data collection	Only official Mufulira Municipal Council digital sources relevant to the assigned CSC 4792 project scope were used. Rejected CDF project documents were excluded from the CDF project dataset.
Description of data collection	Records were extracted from CDF community-project documents, municipal budgets for 2023-2026, approved skills-development bursary documents for 2025-2026, and the Mufulira Integrated Development Plan.
Data source location	Mufulira Municipal Council, Zambia; official website and council-hosted PDF documents.
Data accessibility	Kaggle repository: https://www.kaggle.com/datasets/mambwelchilebela/cs4792-mufulira-municipal-council-datasetgroup-8
Related code	GitHub repository: https://github.com/Mambwe-LC/db-unza26-csc4792-Mufulira-Municipal-Council-group-8.git

Primary council website: <https://www.mufuliracouncil.gov.zm/> | CDF source page used in the extraction workflow: https://www.mufuliracouncil.gov.zm/?page_id=792

Value of the Data

- The data consolidate records that are otherwise distributed across multiple council webpages and PDF documents into machine-readable CSV files.
- The data can support analysis of CDF project distribution, municipal revenue and expenditure planning, ward-level development needs, and planned capital investments.
- Researchers, students, civic organisations, journalists, and policy analysts can reuse the data for local-government transparency, development monitoring, and comparative analysis across councils.
- The coexistence of raw budget-table data and cleaned budget/revenue data supports reproducibility and allows users to inspect how structured records were derived.
- The multi-source structure makes the collection useful for data integration, data quality assessment, text mining, and other CSC 4792 data-mining tasks.

Data Description

The collection is organised into eight pipe-delimited CSV datasets. The datasets preserve source provenance through source-file, source-title, or source-URL fields where those fields were defined during extraction. Table 1 summarises the final dataset components described by the extraction notebook.

Dataset file	Content	Main fields / description
db-unza26-csc4792-mufulira_budget_revenue_2023_2026.csv	Cleaned budget and revenue records, 2023-2026	record_id, year, data_type, category, revenue_source, budget_amount, budget_status, description, source_title, source_url, source_type.
db-unza26-csc4792-mufulira_cdf_skills_applicants.csv	CDF skills-development bursary records	Constituency, record number, ward, zone, pupil/applicant fields, vulnerability, programme, study type, course duration, institution, fees, approval indicator, source file. Direct identifiers require privacy review before public release.
db-unza26-csc4792-mufulira_health_facilities.csv	Health facilities	Facility ID, facility name, facility type, ward,

		ownership, operational status, catchment population, constituency.
db-unza26-csc4792-mufulira_ward_public_consultation_issues.csv	Ward public-consultation issues	Ward name, challenges, solutions, sector, constituency.
db-unza26-csc4792-mufulira_master_capital_investment_framework.csv	Master capital investment framework	Sector, main objective, specific objective/program, project description, cost in ZMW, proposed source of funds, location, ward, responsible agency, implementation period.

1. CDF community-project data

CDF project information was obtained from the official council CDF page, which was used to identify available Community Projects PDF documents for Kantanshi, Kankoyo, and Mufulira Central. Rejected-project documents were excluded. The structured project dataset contains geographic, project, sector, timing, cost, approval, and justification fields together with source-file provenance.

2. Budget and revenue data

Municipal budget data were extracted from official budget documents for 2023, 2024, 2025, and 2026. Two outputs were retained: a raw table-level dataset and a cleaned analytical dataset. The cleaned dataset classifies records into categories such as fees and charges, local taxes/rates, Local Government Equalisation Fund (LGEF), and expenditure.

3. Skills-development bursary data

Skills-development bursary records were extracted from approved-applicant documents for Kantanshi, Kankoyo, and Mufulira Central for 2025 and 2026, including replacement-applicant documents where applicable. Because the source documents may contain direct personal identifiers, the public-release version requires de-identification before publication on Kaggle.

4. Governance and IDP data

Four datasets were extracted from the official Integrated Development Plan: administrative wards and demographics, health facilities, ward public-consultation issues, and the Master Capital Investment Framework. For these tables, the extraction workflow preserved raw cell values and did not apply duplicate removal, value standardisation, generated identifiers, or manual record creation.

Experimental Design, Materials and Methods

The data-acquisition workflow was designed around the format of each source document. Digital tables were extracted directly, while scanned PDF pages were processed through optical character recognition (OCR). All pipelines ended with Pandas DataFrames and pipe-delimited CSV exports.

1. CDF community-project extraction

1. Access the Mufulira Municipal Council CDF webpage.
2. Discover Community Projects PDF links using Requests and BeautifulSoup.
3. Download each PDF directly into memory without requiring a local source-PDF copy.
4. Convert scanned PDF pages to images using pdf2image.
5. Apply OCR using Tesseract through pytesseract.
6. Use regular expressions and predefined ward/sector lists to identify project information.
7. Combine the extracted projects into a Pandas DataFrame.
8. Export the final data as a pipe-delimited CSV file.

2. Budget and revenue extraction

9. Download the four official budget PDFs (2023-2026) using Requests.
10. Process the PDF content in memory.
11. Extract tables using pdfplumber.
12. Preserve extracted tables in a raw dataset.
13. Identify relevant budget and revenue records.
14. Classify records into categories including fees and charges, local taxes/rates, LGEF, and expenditure.
15. Extract revenue sources, budget amounts, and budget statuses.
16. Combine the records in a Pandas DataFrame.
17. Export both raw and cleaned datasets using the pipe delimiter.

3. Skills-development bursary extraction

18. Download the official skills-development PDF documents directly from their URLs using Requests.
19. Process each PDF in memory.
20. Detect whether the PDF contains extractable digital text.
21. For digital PDFs, extract tables using pdfplumber.
22. For scanned PDFs, convert pages to images using pdf2image.
23. Apply OCR using Tesseract/pytesseract.
24. Identify applicant rows and map values to the predefined skills-development fields.
25. Separate guardian/parent names and phone numbers into distinct extracted fields.
26. Combine records from all included constituencies and years in a Pandas DataFrame.
27. Remove duplicate applicant records where applicable.
28. Export the structured dataset using the pipe delimiter.

4. Governance and IDP extraction

29. Download the official Mufulira Municipal Council IDP PDF directly from its URL using Requests.
30. Process the PDF in memory.
31. Extract structured tables using pdfplumber.
32. Detect relevant tables using headers and table structure.
33. Extract raw cell values without cleaning or standardising the values.
34. Preserve the extracted records in Pandas DataFrames.
35. Export each extracted table as a pipe-delimited CSV file.

5. Software and tools

Tool	Purpose
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Python	Data extraction and processing
Requests	Retrieving webpages and PDF documents
BeautifulSoup	Discovering CDF PDF links from HTML
pdf2image	Converting scanned PDF pages to images
Tesseract / pytesseract	OCR of scanned CDF and skills-development records
pdfplumber	Extracting tables from digital PDFs
Regular expressions	Identifying and extracting structured values from OCR/text
Pandas	Structuring records, removing duplicates where applicable, and exporting CSV files

6. Overall workflow

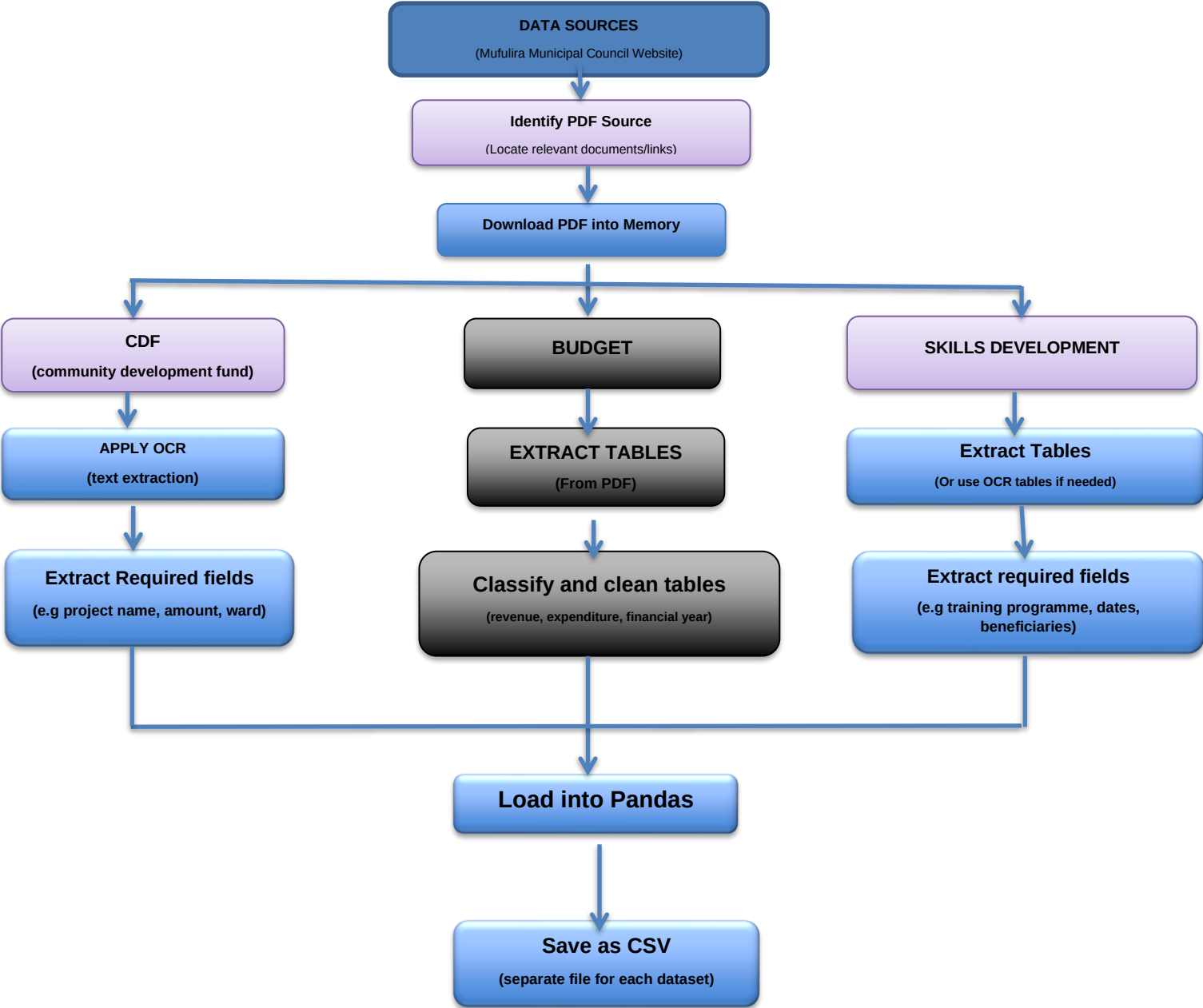
CDF: website -> PDF links -> PDF in memory -> OCR -> field extraction -> Pandas -> CSV

Budget: PDF URLs -> PDF in memory -> table extraction -> classification/cleaning -> Pandas -> CSV

Skills development: PDF URLs -> PDF in memory -> digital table extraction or OCR -> field extraction -> Pandas -> CSV

Governance/IDP: PDF URL -> PDF in memory -> table extraction -> raw structured records -> Pandas -> CSV

Figure showing Data Extraction Workflow for Mufulira Municipal Council Documents.



7. File formatting and naming

All final data files are intended to use the pipe character (|) as the delimiter. For the coursework submission, final filenames should follow the required pattern db-unza26-csc4792-[DESCRIPTION].csv. Any working budget files that currently use an underscore immediately after 'csc4792' should be renamed before final submission so that the naming convention is consistent across the dataset.

Limitations

- The data are limited to information made publicly available through the Mufulira Municipal Council website and linked official PDF documents.
- OCR quality can be affected by scan resolution, page layout, skew, or unusual typography; OCR-derived fields therefore require validation against the source documents.
- The governance/IDP tables were intentionally preserved as raw extracted values, so spelling variants, blank cells, or source-document inconsistencies may remain.
- The collection does not claim to represent unpublished council records or information that was unavailable on the website at the time of extraction.
- Record counts and temporal coverage may change if the council updates or replaces source documents.

Ethics and Privacy Statement

The extraction used publicly accessible council documents and did not involve direct interaction with human participants. However, some skills-development bursary source documents contain fields that can directly identify individuals, including names, National Registration Card (NRC) numbers, dates of birth, guardian/parent names, and telephone numbers. Public availability of a source document should not be treated as permission for unrestricted republication of direct identifiers. Before the dataset is uploaded to a public repository such as Kaggle, direct identifiers should be removed, masked, or otherwise de-identified, while retaining only the minimum non-identifying variables needed for the coursework analysis.

CRedit Author Statement

Complete this section using the five group members' actual contributions. A suggested structure is provided below and should be edited so that it accurately reflects the work performed.

Author	Contributor roles
Kasonkomona Mulenga - 2021519929	Data curation; Investigation; Software; Validation
Ephetred Ndhlovu - 2021463681	Data curation; Investigation; Software; Validation
Kalebalika Chileshe - 2021452344	Data curation; Investigation; Software; Validation
Moses Kaluba - 2021387283	Methodology; Data curation; Validation; Visualization
Mambwe Luke Chilebela - 2021480348	Project administration; Writing - original draft; Writing - review and editing; Data curation

Data Availability

The final curated dataset is intended to be publicly available on Kaggle.

<https://www.kaggle.com/datasets/mambwelchilebela/cs4792-mufulira-municipal-council-datasetgroup-8>

The extraction and preprocessing notebook/codebase is intended to be available on GitHub.

<https://github.com/Mambwe-LC/db-unza26-csc4792-Mufulira-Municipal-Council-group-8.git>

Acknowledgements

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work described in this data article.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During preparation of this manuscript, the authors used OpenAI ChatGPT to assist with manuscript organisation and language refinement. The authors are responsible for reviewing, verifying, and editing the content and take full responsibility for the final submitted version.

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